

MCA Level Notes: Big Data and Analytics (6-Marks Answers)

Q1: Difference between Reporting and Analysis in Analytics and Big Data

Reporting:

- Reporting refers to the process of organizing data into informational summaries.
- It focuses on what *has happened* in the past.
- Tools: Dashboards, static reports, visual charts.
- Example: Monthly sales report showing revenue by region.

Analysis:

- Analysis goes deeper to interpret the data, find patterns, correlations, and trends.
- It answers *why* something happened and *what could happen*.
- Tools: Data mining, predictive models.
- Example: Analyzing sales drop using correlation between ad spending and customer response.

Difference Table:

Feature	Reporting	Analysis
Focus	Descriptive	Diagnostic/Prescriptive
Output	Dashboards, Charts	Models, Insights
Tools	BI Tools, SQL	ML Models, Python, R
Time-frame	Historical	Historical + Future

Q2: Types of Analytics with Example

1. Descriptive Analytics:

2. Summarizes past data.

3. *Example:* Retail company analyzing sales from last year.

4. Diagnostic Analytics:

5. Finds the cause behind an event.

6. *Example:* Investigating why sales dropped in Q2.

7. Predictive Analytics:

8. Forecasts future outcomes using data trends.
9. *Example:* Predicting next month's churn rate in a telecom company.
10. **Prescriptive Analytics:**
11. Suggests actions to affect desired outcomes.
12. *Example:* Recommending discounts to improve customer retention.

Diagram:

Data -> Descriptive -> Diagnostic -> Predictive -> Prescriptive

Q3: Steps to Develop an Analytics Team in a New Organization

1. **Define Goals:** Understand business needs.
2. **Hire Key Roles:** Data engineers, analysts, data scientists.
3. **Infrastructure Setup:** Choose tools and platforms (Hadoop, Spark, cloud tools).
4. **Data Governance:** Set up security, compliance, and quality control.
5. **Process Design:** Establish workflow for data ingestion, transformation, visualization.
6. **Training & Culture:** Develop data literacy among departments.
7. **Iterative Improvement:** Review and upgrade the team structure regularly.

Q4: What is Text Analytics and How It Is Useful?

Definition: Text analytics is the process of extracting useful insights and patterns from unstructured text data using NLP and machine learning techniques.

Uses:

- Sentiment analysis (e.g., social media feedback).
- Topic extraction from articles.
- Document classification (spam detection).
- Customer feedback summarization.

Benefits:

- Understand customer sentiment.
 - Monitor brand reputation.
 - Automate content categorization.
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Q5: What is Apache Pig and How it Differs from Traditional Database?

Apache Pig:

- High-level platform for creating MapReduce programs for big data processing.
- Uses *Pig Latin* language.

Differences with Traditional DBs:

Feature	Apache Pig	Traditional DB
Language	Pig Latin	SQL
Schema	Optional	Strict
Processing	Batch (MapReduce)	Transactional
Storage	HDFS	Disk-based RDBMS
Use Case	Big Data	Structured Data

Q6: Execution Modes of Pig and Its Components

Execution Modes:

1. **Local Mode:**
 2. All data and scripts run on a single JVM.
 3. Used for testing.
4. **MapReduce Mode:**
 5. Data is processed on a Hadoop cluster.
 6. Suitable for large datasets.

Basic Components:

1. **Grunt Shell:** Interactive shell for executing Pig Latin commands.
 2. **Pig Latin:** Scripting language.
 3. **Execution Engine:** Converts scripts to MapReduce jobs.
 4. **UDFs (User Defined Functions):** For custom processing.
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Q7: Components of Text Analytics Pipeline

1. **Text Collection:** Collect unstructured data from sources (emails, social media).
2. **Text Preprocessing:**
3. Tokenization
4. Stop word removal
5. Lemmatization/Stemming
6. **Feature Extraction:**
7. TF-IDF, Word embeddings (Word2Vec).
8. **Text Classification/Clustering:**
9. Group or categorize text (e.g., spam vs. ham).
10. **Visualization and Insight Extraction:**
11. Charts, sentiment dashboards.

Diagram:

Collection -> Preprocessing -> Feature Extraction -> Modeling -> Insights

Q8: What is Hive? Comparison with Traditional DB

Hive:

- A data warehouse infrastructure on Hadoop.
- Converts SQL-like queries (HiveQL) into MapReduce jobs.

Comparison Table:

Feature	Hive	Traditional DB
Query Lang	HiveQL	SQL
Schema	Flexible	Fixed
Processing	Batch (MapReduce)	Real-time
Speed	Slower	Faster
Use Case	Big Data analytics	OLTP, structured data

Q9: MongoDB vs SQL for Unstructured Data

MongoDB:

- NoSQL document store.
- Uses BSON documents.

Advantages for Unstructured Data:

- Schema-less
- Embedded documents
- Faster for hierarchical or varied data

SQL (RDBMS):

- Fixed schema
- Harder to manage nested/unstructured data

Example:

- MongoDB:

```
{ name: "Alice", reviews: ["good", "bad"] }
```

- SQL:

```
CREATE TABLE Reviews (user_id INT, review TEXT);
```

Q10: RDBMS vs MongoDB Data Types

RDBMS Type	MongoDB Type
INT	NumberInt
VARCHAR	String
DATE	Date
BLOB	BinData
BOOLEAN	Boolean
ARRAY (via JOIN)	Embedded Array

Q11: Stages of Data Analysis for Decision Points

1. Data Collection

2. Source decision: What data and from where?

3. Data Cleaning

4. Identify outliers, missing values.

5. Data Exploration

6. Visualize trends, identify hypotheses.

7. Modeling/Analysis

8. Decision: What technique to use (Regression, Clustering)?

9. Interpretation

10. Decision: What do results imply?

11. Deployment

12. Apply model in real-world system.

Q12: Key Concepts and Components of HBase

Key Concepts:

- Column-oriented NoSQL database on Hadoop.
- Stores large tables across multiple machines.

Components:

1. **HMaster:** Controls region assignment.
 2. **RegionServer:** Manages read/write for regions.
 3. **Zookeeper:** Coordinates and manages services.
 4. **HFile:** Physical storage format.
 5. **MemStore:** In-memory store before flush to disk.
 6. **Write-Ahead Log (WAL):** Ensures durability.
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Q13: Suitability of Hive for Analytical Queries

Justification:

- Hive is designed for large-scale batch processing.
- Supports complex joins, aggregations.
- Easily integrates with Hadoop ecosystem.

Features Supporting Analytics:

- HiveQL similar to SQL: low learning curve.
- Partitioning improves performance.
- UDFs for advanced computation.
- Integration with BI tools.

Conclusion: Hive is ideal for analytical queries on big structured data, but not suitable for real-time operations.

Q14: MapReduce Architecture

Definition: MapReduce is a programming model used for processing large data sets with a distributed algorithm on a Hadoop cluster.

Components:

1. **Map Function:**
2. Takes input data and converts it into a set of key-value pairs.
3. **Shuffle and Sort:**
4. Intermediate key-value pairs are grouped and sorted.
5. **Reduce Function:**
6. Aggregates values for each key and produces output.

Architecture Diagram:

```
graph TD
    A[Input Splits] --> B[Map Phase]
    B --> C["Intermediate <key, value>"]
    C --> D[Shuffle & Sort]
    D --> E[Reduce Phase]
    E --> F[Final Output]
```

Example:

- Word count problem:
- Map: <"cat cat dog"> -> {cat:1, cat:1, dog:1}
- Reduce: {cat:[1,1]} -> {cat:2}, {dog:[1]} -> {dog:1}

Advantages:

- Scalable and fault-tolerant.
- Handles very large datasets.

Limitations:

- Not suitable for low-latency queries.
- Requires writing custom logic for even simple tasks.

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